

I. EXPERIMENTAL SETUP AND RESULTS

The experiments evaluate deterministic short-term vessel trajectory forecasting on the compact EnvShip-Bench paper benchmark. The goal is to predict the next 10 minutes of vessel motion from the previous 10 minutes under a unified 30→30 forecasting protocol. All models are evaluated on the same curated compact split family so that architectural differences, rather than dataset-version differences, explain performance differences.

The evaluation reported here focuses on coordinate-only deterministic forecasting. We compare five recurrent baselines, namely Seq2Seq, GRU, Bi-GRU, LSTM, and Bi-LSTM, against a single invertible forecasting model, Kine-Real-NVP. The Kine-Real-NVP model corresponds to the KineRoute-NVP implementation used in this repository, but the run name in the results section is written as Kine-Real-NVP for easier comparison with the baseline family.

A. Execution Environment

Experiments were run on an NVIDIA GeForce RTX 2050 GPU with 4 GB of memory in the local `data_science` Python environment. Training used PyTorch with CUDA execution. For reproducibility, every compact-benchmark run was launched from a checked-in experiment YAML and wrote a fully resolved configuration, command record, per-epoch metric log, checkpoints, and final machine-readable metrics into a timestamped results directory under `results/`.

B. Forecasting Protocol

All benchmark samples follow the same short-term maritime forecasting protocol:

- observation horizon: 10 minutes,
- prediction horizon: 10 minutes,
- sampling interval: 20 seconds,
- history length: 30 points,
- future length: 30 points.

Each trajectory is represented in vessel-centric local planar coordinates measured in meters, with the last observed point acting as the anchor. Deterministic forecasting is evaluated with Average Displacement Error (ADE) and Final Displacement Error (FDE), both in meters.

C. Datasets

The experiments use the official EnvShip-Bench release on Hugging Face published by `mark000071`. We intentionally use the compact paper subsets rather than the later cross-domain v2 release. Specifically, we combine the DMA mini benchmark subset `clean_ship_core_lite_v1` with the NOAA mini benchmark subset `clean_ship_core_lite_v1`.

Within the repository, these two subsets are exposed under `data/envship/` through the local aliases `paper_dma_clean_ship_core_lite_v1` and `paper_noaa_clean_ship_core_lite_v1`.

The resulting combined split sizes are:

$$N_{\text{train}} = 29635, \quad N_{\text{val}} = 3667, \quad N_{\text{test}} = 3262.$$

The DMA compact subset contributes 19,500 training windows and the NOAA compact subset contributes 10,135 training windows, giving a combined compact benchmark that is closely aligned with the paper’s controlled experimental setting.

D. Dataset-Specific Preprocessing

Each CSV row is one fixed-length vessel-trajectory sample containing 30 historical points and 30 future points. The preprocessing pipeline parses the `hist_x_json`, `hist_y_json`, `fut_x_json`, and `fut_y_json` arrays and stores them in compressed NumPy artifacts for training.

For the recurrent baselines, the raw coordinate history and future arrays are used directly, followed by normalization computed from the training split only. For Kine-Real-NVP, the compact benchmark is additionally transformed with two geometric preprocessing steps:

- alignment to the last observed heading,
- displacement-chart encoding of the future trajectory.

Heading alignment rotates each sample so that the last observed motion direction is canonicalized. The displacement chart represents the future as stepwise displacements rather than absolute future positions. This substantially reduces directional variance and makes the target representation easier to model on the compact benchmark.

E. Models

1) *Recurrent Baselines*: Five recurrent baselines are evaluated under the same compact-benchmark protocol.

The Seq2Seq model uses a GRU encoder and a GRU decoder with teacher forcing during training. The GRU and Bi-GRU baselines use a recurrent encoder followed by a multilayer perceptron head that predicts the full 30-step future. The LSTM and Bi-LSTM baselines use the same prediction head structure but replace the GRU encoder with an LSTM encoder.

All recurrent baselines use hidden dimension 128 and 2 recurrent layers. Their parameter counts are:

- Seq2Seq: 299,778,
- GRU: 174,012,
- Bi-GRU: 438,460,
- LSTM: 223,932,
- Bi-LSTM: 571,068.

2) *Kine-Real-NVP*: Kine-Real-NVP is an invertible supervised forecasting model built from routed RealNVP-style affine coupling blocks over a kinematic trajectory chart. In the compact-benchmark experiments, the strongest configuration uses:

- 6 routed coupling blocks,
- hidden dimension 256,
- 2 hidden layers per coupling network,
- scale bound 1.5,
- heading-aligned displacement-chart targets,
- pure ADE optimization without auxiliary physics penalties.

The full model has 2,931,408 trainable parameters.

Although the mapping is numerically invertible as a function from history-chart space to future-chart space, the learned inverse is not interpreted as a physically correct recovery of the true past trajectory from the future. The benchmark is a supervised forecasting task rather than a naturally bijective dynamical reconstruction task.

F. Training Protocol

All reported compact-benchmark experiments use 80 epochs, batch size 256, AdamW optimization, learning rate 10^{-3} , and weight decay 10^{-4} . The learning-rate scheduler is ReduceLROnPlateau with factor 0.5 and patience 8. For the canonical Kine-Real-NVP run, the experiment YAML sets the early-stopping patience to 999 so that the full 80-epoch schedule is executed while still preserving the same checkpointing and logging interface as the other runs.

The recurrent baselines are trained with normalized coordinates and mean-squared error in normalized target space. Seq2Seq uses teacher forcing ratio 0.5 during training. Kine-Real-NVP is trained with pure ADE loss, with all auxiliary velocity, acceleration, turn, feasibility, reverse, position-MSE, and extra FDE weights set to zero. This was the strongest Kine-Real-NVP setting on the compact benchmark.

G. Evaluation Metrics

Average Displacement Error is defined as

$$\text{ADE} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \|\hat{y}_{i,t} - y_{i,t}\|_2,$$

where N is the number of samples, $T = 30$ is the forecasting horizon, $y_{i,t}$ is the target future position at step t , and $\hat{y}_{i,t}$ is the predicted future position.

Final Displacement Error is defined as

$$\text{FDE} = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_{i,T} - y_{i,T}\|_2.$$

Both metrics are measured in meters. Lower values are better.

II. RESULTS

Table ?? reports the compact-benchmark results obtained from the unified `results/` artifact tree. The recurrent baselines closely reproduce the paper’s coordinate-only deterministic regime, while Kine-Real-NVP substantially improves on all of them.

The recurrent baselines all moved into the paper regime once the correct compact benchmark was used. Seq2Seq was reproduced almost exactly. GRU, Bi-GRU, LSTM, and Bi-LSTM all slightly outperformed the paper reference values in this implementation. Kine-Real-NVP achieved the strongest performance overall, with a test result of 50.40 ADE and 112.09 FDE.

The Kine-Real-NVP epoch trajectory is also notable: the best validation result occurred at epoch 36, well before

the 80-epoch schedule completed. This indicates that the compact benchmark is highly compatible with the aligned displacement-chart parameterization and that the model reaches a strong forecasting regime early in training.

Finally, the invertibility diagnostics should be interpreted carefully. The model remained numerically invertible with small reconstruction error in latent chart space, but reverse-history prediction error remained very large. This is not a contradiction. The learned mapping is invertible as a parameterized function, but the benchmark task itself does not imply that a true vessel future uniquely determines the true vessel past in a semantically meaningful one-to-one way.

TABLE I

COMPACT BENCHMARK RESULTS ON THE COORDINATE-ONLY DETERMINISTIC SETTING. PAPER VALUES ARE LISTED FOR REFERENCE, AND REPOSITORY RESULTS REPORT THE COMPLETED COMPACT-BENCHMARK RUNS FROM THIS WORK.

Model	Paper ADE	Paper FDE	Best Val ADE	Best Val FDE	Test ADE	Test FDE
Seq2Seq	59.57	131.43	59.62	128.11	60.19	129.41
GRU	59.72	133.69	57.65	123.44	58.13	125.47
Bi-GRU	61.08	134.51	55.35	119.58	57.44	124.64
LSTM	65.72	156.92	59.22	125.94	59.58	127.22
Bi-LSTM	74.97	193.37	57.81	123.87	58.85	126.98
Kine-Real-NVP	–	–	50.26	110.81	50.40	112.09